

**THE HERITAGE SCHOOL**  
**FIRST TERM EXAMINATION, 2026-27**  
**PHYSICS PAPER 1 (THEORY) · CLASS 12 · PRACTICE PAPER 3**

**Maximum Marks: 70**

**Time Allotted: Three Hours**

**Instructions to Candidates**

- You are allowed an additional fifteen minutes for only reading the question paper. You must NOT start writing during this time.
- There are **twenty questions** in this paper. **Answer all questions.**
- There are **four sections** in the paper: A, B, C and D. Internal choices have been provided in **two** questions each in Sections B, C and D.
- **Section A** consists of one question having fourteen sub-parts of **1 mark** each.
- While attempting the multiple choice sub-parts you are required to write only ONE option as the answer.
- **Section B** consists of seven questions of **2 marks** each.
- **Section C** consists of nine questions of **3 marks** each.
- **Section D** consists of three questions of **5 marks** each.
- The intended marks for questions or parts of questions are given in brackets [ ].
- All working, including rough work, should be done on the same sheet as, and adjacent to, the rest of the answer.
- A list of useful physical constants is given at the end of this paper.
- A simple scientific calculator without a programmable memory may be used for calculations.

*Syllabus covered:* Electrostatics; Current Electricity; Magnetic Effect of Current and Magnetism; Electromagnetic Induction and Alternating Current; Electromagnetic Waves; Optics (Ray Optics and Optical Instruments, and Wave Optics).

*Not examined in this paper:* Dual Nature of Radiation and Matter, Atoms and Nuclei, Electronic Devices. Polarisation is no longer in the Wave Optics syllabus and is not examined here either.

**SECTION A - 14 MARKS**

**Question 1 [14 × 1]**

In sub-parts (i) to (vii) choose the correct alternative. Answer sub-parts (viii) to (xiv) briefly.

- (i) The work done in moving a charge of  $2.0 \mu\text{C}$  from a point at a potential of  $10 \text{ V}$  to a point at a potential of  $40 \text{ V}$  is:
- (a)  $20 \mu\text{J}$   
(b)  $60 \mu\text{J}$   
(c)  $80 \mu\text{J}$   
(d)  $100 \mu\text{J}$
- (ii)  $n$  identical resistors give a total resistance  $R_s$  in series and  $R_p$  in parallel. The ratio  $R_s : R_p$  is:
- (a)  $n : 1$   
(b)  $1 : n$   
(c)  $n^2 : 1$   
(d)  $1 : n^2$
- (iii) A current-carrying rectangular loop is placed in a uniform magnetic field with the **plane of the loop parallel** to the field. The torque on it is:
- (a) zero.  
(b) minimum but not zero.  
(c) maximum.  
(d) half of its maximum value.

- (iv) A step-up transformer has  $N_s : N_p = 5 : 1$ . If the primary current is 10 A, the secondary current in the ideal case is:
- 50 A
  - 10 A
  - 5 A
  - 2 A
- (v) A convex lens of power +6 D is placed in contact with a concave lens of power –2 D. The focal length of the combination is:
- 12.5 cm
  - 25 cm
  - 25 cm
  - 50 cm
- (vi) In an interference pattern the path difference at a point is  $1.5\lambda$ . The point is:
- bright.
  - dark.
  - of half the maximum intensity.
  - the centre of the pattern.
- (vii) **Assertion:** a voltmeter is connected in parallel with the component whose potential difference is to be measured. **Reason:** a voltmeter has a very high resistance, so it draws almost no current from the circuit.
- Both Assertion and Reason are true and Reason is the correct explanation for Assertion.
  - Both Assertion and Reason are true but Reason is not the correct explanation for Assertion.
  - Assertion is true and Reason is false.
  - Assertion is false and Reason is true.
- (viii) State the SI unit of electric flux.
- (ix) Why is a potentiometer preferred to a voltmeter for measuring the emf of a cell?
- (x) Define the magnetic dipole moment of a current-carrying coil.
- (xi) Name the rule used to find the direction of the current induced in a straight conductor moving in a magnetic field.
- (xii) Which electromagnetic radiation has photons of the greatest energy?
- (xiii) Write the relation between the focal length and the radius of curvature of a spherical mirror.
- (xiv) What is the shape of the wavefront reaching the earth from a source at a very great distance?

## SECTION B - 14 MARKS

### Question 2 [2]

The capacitance of a parallel plate air capacitor is 8.0 pF and the plates are 2.0 mm apart. Calculate the area of each plate.

### Question 3 [2]

In a metre bridge the balance point is obtained at 45 cm from the left end when a 5.0  $\Omega$  resistance is in the left gap. Calculate the unknown resistance in the right gap, and state one precaution to be taken.

### Question 4 [2]

State any two ways of increasing the current sensitivity of a moving coil galvanometer.

**OR**

Two long parallel wires 4.0 cm apart carry currents of 3.0 A and 6.0 A in the **same** direction. Calculate the force per unit length between them and state whether it is attractive or repulsive.

### Question 5 [2]

State any two applications of eddy currents.

**Question 6 [2]**

- (i) An electromagnetic wave has a wavelength of about  $10^{-10}$  m. Name it.  
(ii) State one use of gamma rays.

**Question 7 [2]**

Why is a convex mirror preferred to a plane mirror as the rear-view mirror of a vehicle?

**OR**

An object is placed 10 cm in front of a concave mirror of focal length 15 cm. Find the position and nature of the image.

**Question 8 [2]**

Define coherent sources of light and state how two coherent sources are obtained in Young's double slit experiment.

**SECTION C - 27 MARKS****Question 9 [3]**

In Young's double slit experiment, derive an expression for the fringe width. State the two conditions used.

**Question 10 [3]**

Derive the expression  $\rho = m/ne^2\tau$  for the resistivity of a conductor, and hence explain why the resistivity of a metal increases with temperature.

**OR**

Two cells of emf 18 V and 16 V, each of internal resistance  $1.0\ \Omega$ , are connected in parallel with each other, and the combination sends current through an external resistance of  $4.0\ \Omega$ . Using Kirchhoff's laws, calculate the current through the  $4.0\ \Omega$  resistor.

**Question 11 [3]**

State the Biot-Savart law and use it to find the magnetic flux density at the centre of a circular coil of  $N$  turns, radius  $R$ , carrying a current  $I$ .

**Question 12 [3]**

With a labelled diagram, explain the construction and working of a moving coil galvanometer. Why is a radial magnetic field used?

**Question 13 [3]**

Derive an expression for the root mean square value of an alternating current  $I = I_0 \sin \omega t$ , and state why it, rather than the mean value, is quoted for a.c.

**Question 14 [3]**

Derive an expression for the mutual inductance of two long coaxial solenoids of the same length  $l$ , with  $N_1$  and  $N_2$  turns and cross-sectional area  $A$  of the inner one.

**Question 15 [3]**

Draw a ray diagram for a simple microscope forming its image at the least distance of distinct vision, write its magnifying power, and state the effect of using a lens of smaller focal length.

**Question 16 [3]**

A prism of refracting angle  $60^\circ$  produces a minimum deviation of  $30^\circ$ . Calculate the refractive index of its material and the speed of light inside it.

**Question 17 [3]**

State any three points of difference between the interference pattern of two narrow slits and the diffraction pattern of a single slit.

**OR**

In a single-slit diffraction experiment the first minimum for light of wavelength 500 nm occurs at  $30^\circ$ . Calculate the width of the slit and the angular width of the central maximum.

### SECTION D - 15 MARKS

#### Question 18 [5]

- (a) Using Gauss' theorem, derive an expression for the electric field intensity at a point outside a uniformly charged thin spherical shell. State the field inside it. [3]
- (b) A shell of radius 10 cm carries a charge of  $2.0 \mu\text{C}$ . Calculate the field at 20 cm and at 5.0 cm from its centre. [2]

OR

- (a) Using Gauss' theorem, derive an expression for the electric field intensity due to an infinitely long straight uniformly charged wire of linear charge density  $\lambda$ . [3]
- (b) Capacitors of  $3.0 \mu\text{F}$  and  $6.0 \mu\text{F}$  are connected across a 60 V supply, first in series and then in parallel. Calculate the total energy stored in each case. [2]

#### Question 19 [5]

- (a) Derive the mirror formula  $1/v + 1/u = 1/f$  for a concave mirror forming a real image, with a labelled ray diagram. [3]
- (b) An object 4.0 cm high is placed 30 cm from a concave mirror of focal length 20 cm. Find the position, size and nature of the image. [2]

OR

- (a) Derive an expression for the equivalent focal length of two thin lenses of focal lengths  $f_1$  and  $f_2$  placed in contact. [3]
- (b) A convex lens of focal length 30 cm is placed in contact with a concave lens of focal length 20 cm. Find the focal length and nature of the combination. [2]

#### Question 20 [5]

Read the passage and answer the questions that follow.

A domestic circuit in India is supplied at 220 V, 50 Hz. Appliances are wired in parallel, so that each gets the full supply voltage and can be switched independently. The live wire carries a fuse – a short piece of wire of low melting point which melts and breaks the circuit if the current exceeds its rating. An earth wire connects the metal body of an appliance to a plate buried in the ground, so that a leakage of current to the body flows harmlessly to earth instead of through the user.

- (i) Why are domestic appliances connected in parallel and not in series?
- (ii) An electric iron is rated 2000 W, 220 V. Calculate the current it draws.
- (iii) Calculate its resistance when working.
- (iv) Would a 5 A fuse be suitable for this iron? Give a reason.
- (v) State why the fuse is placed in the live wire rather than the neutral wire.

## USEFUL CONSTANTS AND RELATIONS

|     | Quantity                                   | Symbol             | Value                                        |
|-----|--------------------------------------------|--------------------|----------------------------------------------|
| 1.  | Permittivity of vacuum                     | $\epsilon_0$       | $8.85 \times 10^{-12} \text{ F m}^{-1}$      |
| 2.  | Constant for Coulomb's law                 | $1/4\pi\epsilon_0$ | $9 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$ |
| 3.  | Permeability of vacuum                     | $\mu_0$            | $4\pi \times 10^{-7} \text{ H m}^{-1}$       |
| 4.  | Constant in magnetism                      | $\mu_0/4\pi$       | $1 \times 10^{-7} \text{ H m}^{-1}$          |
| 5.  | Charge of a proton = charge of an electron | $e$                | $1.6 \times 10^{-19} \text{ C}$              |
| 6.  | Mass of an electron                        | $m_e$              | $9.1 \times 10^{-31} \text{ kg}$             |
| 7.  | Mass of a proton                           | $m_p$              | $1.67 \times 10^{-27} \text{ kg}$            |
| 8.  | Speed of light in vacuum                   | $c$                | $3 \times 10^8 \text{ m s}^{-1}$             |
| 9.  | Acceleration due to gravity                | $g$                | $9.8 \text{ m s}^{-2}$                       |
| 10. | Least distance of distinct vision          | $D$                | 25 cm                                        |

## ANSWER KEY & MARKING NOTES

Practice Paper 3. **Do not read this section before attempting the paper.** Every numerical answer was recomputed independently; the notes give the decisive step rather than a full solution.

### Section A

| Q        | Answer                                                                                                                                                                                                                                                                 |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 (i)    | <b>(b)</b> $W = q(V_B - V_A) = (2.0 \times 10^{-6})(30) = 6.0 \times 10^{-5} \text{ J} = \mathbf{60 \mu J}$ .                                                                                                                                                          |
| 1 (ii)   | <b>(c)</b> $R_s = nR$ and $R_p = R/n$ , so the ratio is $n^2 : 1$ .                                                                                                                                                                                                    |
| 1 (iii)  | <b>(c)</b> Maximum. $\tau = NIAB \sin \theta$ with $\theta$ the angle between the <b>normal</b> and <b>B</b> . Plane parallel to <b>B</b> means the normal is <b>perpendicular</b> to it, $\theta = 90^\circ$ , $\sin \theta = 1$ .                                    |
| 1 (iv)   | <b>(d)</b> 2 A. Power is conserved, so $I_s/I_p = N_p/N_s = 1/5$ ; a step-up transformer steps the current <b>down</b> .                                                                                                                                               |
| 1 (v)    | <b>(b)</b> $P = 6 - 2 = +4 \text{ D}$ , so $f = 1/4 \text{ m} = \mathbf{+25 \text{ cm}}$ (converging).                                                                                                                                                                 |
| 1 (vi)   | <b>(b)</b> Dark. $1.5\lambda = (2n-1)\lambda/2$ with $n = 2$ , an <b>odd</b> multiple of $\lambda/2$ , so the two waves arrive in antiphase.                                                                                                                           |
| 1 (vii)  | <b>(a)</b> Both true, and the high resistance is precisely why a parallel connection does not disturb the circuit it measures.                                                                                                                                         |
| 1 (viii) | <b>N m<sup>2</sup> C<sup>-1</sup></b> (equivalently V m).                                                                                                                                                                                                              |
| 1 (ix)   | At balance <b>no current is drawn</b> from the cell, so there is no drop across its internal resistance and the true emf is measured. A voltmeter always draws some current and therefore reads the <b>terminal potential difference</b> , which is less than the emf. |
| 1 (x)    | It is the product of the current and the area of the coil, $m = NIA$ , directed along the normal to the plane of the coil as given by the right-hand rule. SI unit: A m <sup>2</sup> .                                                                                 |
| 1 (xi)   | <b>Fleming's right hand rule.</b>                                                                                                                                                                                                                                      |
| 1 (xii)  | <b>Gamma rays</b> – they have the shortest wavelength and hence the highest frequency, and $E = h\nu$ .                                                                                                                                                                |
| 1 (xiii) | <b>f = R/2</b> (valid for paraxial rays).                                                                                                                                                                                                                              |
| 1 (xiv)  | <b>Plane.</b> A spherical wavefront of very large radius is, over a small region, indistinguishable from a plane.                                                                                                                                                      |

### Section B

| Q      | Answer                                                                                                                                                                                                                                                                                                                                                              |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2      | $A = Cd/\epsilon_0 = (8.0 \times 10^{-12})(2.0 \times 10^{-3})/8.85 \times 10^{-12} = \mathbf{1.8 \times 10^{-3} \text{ m}^2}$ (about 18 cm <sup>2</sup> ).                                                                                                                                                                                                         |
| 3      | $R/S = l/(100-l)$ gives $S = 5.0 \times 55/45 = \mathbf{6.1 \Omega}$ . Precaution (any one): keep the null point near the middle of the wire so that the percentage error is least; press the jockey gently and vertically so as not to change the wire's cross-section; use a series resistance to protect the galvanometer while finding the approximate balance. |
| 4      | Current sensitivity = $NAB/C$ , so any two of: increase the <b>number of turns N</b> ; increase the <b>area A</b> of the coil; use a <b>stronger field B</b> (stronger magnet, soft-iron core); use a suspension of <b>smaller torsion constant C</b> (a fine phosphor-bronze strip).                                                                               |
| 4 (OR) | $F/I = \mu_0 I_1 I_2 / 2\pi r = (2 \times 10^{-7})(3.0)(6.0)/0.040 = \mathbf{9.0 \times 10^{-5} \text{ N m}^{-1}}$ , and it is <b>attractive</b> because the currents are parallel.                                                                                                                                                                                 |
| 5      | Any two: electromagnetic <b>damping</b> in moving coil galvanometers and in dead-beat instruments; <b>induction furnaces</b> for melting metals; <b>electromagnetic brakes</b> in trains; the <b>speedometer</b> and induction-type energy meters; induction cooking hobs.                                                                                          |
| 6      | (i) <b>X-rays</b> (0.1 nm is the X-ray region; it is also the order of atomic spacings, which is why X-rays diffract from crystals). (ii) Any one: radiotherapy for the treatment of cancer; sterilising surgical instruments and food; industrial radiography to detect flaws in castings and welds.                                                               |
| 7      | A convex mirror always gives an <b>erect, virtual, diminished</b> image, and so has a much <b>wider field of view</b> than a plane mirror of the same size; the driver can see a longer stretch of the road behind. (The cost is that distances are underestimated.)                                                                                                |

| Q      | Answer                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7 (OR) | $1/v = 1/f - 1/u = 1/(-15) - 1/(-10) = +1/30$ , so $v = +30$ cm. $m = -v/u = -30/(-10) = +3$ . The image is <b>virtual, erect and three times magnified</b> , 30 cm behind the mirror – as expected for an object <b>inside</b> the focus.                                                                                                                                                                               |
| 8      | Two sources are <b>coherent</b> if they emit light of the same frequency with a <b>constant phase difference</b> between them. In Young's experiment a single narrow slit S is illuminated first, and the two slits $S_1$ and $S_2$ are illuminated by <b>the same wavefront</b> from S; being <b>derived from one source</b> , any phase change in S occurs in both together, so their phase difference stays constant. |

## Section C

| Q       | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9       | Path difference at P, a distance $y$ from the centre, is $\Delta = yd/D$ – obtained by $S_2P^2 - S_1P^2 = 2yd$ and dividing by $(S_2P + S_1P) \approx 2D$ . <b>[1]</b> Bright fringes need $\Delta = n\lambda$ , so $y_n = n\lambda D/d$ . <b>[1]</b> The fringe width is the spacing of consecutive fringes, $\beta = y_{n+1} - y_n = \lambda D/d$ , the same for bright and dark fringes, so the pattern is evenly spaced. <b>[1]</b> Conditions used: $d \ll D$ and $y \ll D$ (small angles), and the sources must be coherent and of equal intensity for good contrast.                                                                                                                                                                                                                                                                                                               |
| 10      | In a field $E$ an electron gains $v_d = eE\tau/m$ between collisions. <b>[1]</b> The current is $I = neAv_d$ , so the current density is $J = ne^2\tau E/m = \sigma E$ – which is Ohm's law in microscopic form, with $\sigma = ne^2\tau/m$ . <b>[1]</b> Hence $\rho = 1/\sigma = m/ne^2\tau$ . <b>[1]</b> As the temperature rises, the lattice ions vibrate with larger amplitude, the electrons collide more often, so the <b>relaxation time <math>\tau</math> falls</b> while $n$ is essentially constant – therefore $\rho$ <b>rises</b> .                                                                                                                                                                                                                                                                                                                                          |
| 10 (OR) | Let $I_1$ and $I_2$ be the currents from the 18 V and 16 V cells; the current in the $4.0 \Omega$ resistor is $I_1 + I_2$ (junction rule). <b>[1]</b> Loop 1: $18 = 1.0I_1 + 4.0(I_1 + I_2)$ . Loop 2: $16 = 1.0I_2 + 4.0(I_1 + I_2)$ . <b>[1]</b> Subtracting: $I_1 - I_2 = 2$ , and adding: $34 = I_1 + I_2 + 8(I_1 + I_2) = 9I$ , so $I = 3.8$ A through the $4.0 \Omega$ resistor ( $I_1 = 2.9$ A, $I_2 = 0.89$ A). <b>[1]</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 11      | <b>Law:</b> for a current element $I d\mathbf{l}$ , $d\mathbf{B} = (\mu_0/4\pi)I d\mathbf{l} \sin \theta/r^2$ , directed along $d\mathbf{l} \times \hat{\mathbf{r}}$ . <b>[1]</b> For a point at the centre, every element is at the same distance $R$ and is <b>perpendicular</b> to the radius, so $\theta = 90^\circ$ and all the $d\mathbf{B}$ point the same way along the axis; they therefore add as scalars. <b>[1]</b> $B = (\mu_0/4\pi R^2) \oint dl = (\mu_0/4\pi R^2)(2\pi R)$ , so for $N$ turns $\mathbf{B} = \mu_0 N \mathbf{I}/2R$ . <b>[1]</b>                                                                                                                                                                                                                                                                                                                           |
| 12      | <b>Construction:</b> a rectangular coil of many turns of fine insulated wire wound on a light aluminium frame, suspended by a phosphor-bronze strip between the concave poles of a strong permanent magnet, with a cylindrical soft-iron core inside it and a mirror or pointer to read the deflection. <b>[1]</b> <b>Working:</b> the current-carrying coil experiences a torque $N\mathbf{I} \times \mathbf{B}$ which turns it until the restoring couple $C\phi$ of the suspension balances it; hence $N\mathbf{I} \times \mathbf{B} = C\phi$ and $\phi \propto I$ – a <b>linear</b> scale. <b>[1]</b> The concave poles and the soft-iron core make the field <b>radial</b> , so the plane of the coil always contains $\mathbf{B}$ and $\sin \theta = 1$ <b>in every position</b> . Without it the torque would vary as $\sin \theta$ and the scale would not be uniform. <b>[1]</b> |
| 13      | The mean of $I^2$ over one cycle is $\langle I^2 \rangle = I_0^2 \langle \sin^2 \omega t \rangle$ . <b>[1]</b> Since $\sin^2 \omega t = (1 - \cos 2\omega t)/2$ and the cosine averages to zero over a full cycle, $\langle \sin^2 \omega t \rangle = 1/2$ . <b>[1]</b> Hence $I_{\text{rms}} = I_0/\sqrt{2} = 0.707 I_0$ . <b>[1]</b> The <b>mean</b> value over a full cycle is <b>zero</b> , so it says nothing useful; the rms value is the steady direct current that would produce the same heating ( $I^2 R$ ) in the same resistor, which is what an a.c. supply actually delivers.                                                                                                                                                                                                                                                                                               |
| 14      | Let a current $I_2$ flow in the outer solenoid. Inside it the field is uniform, $B_2 = \mu_0 n_2 I_2 = \mu_0 (N_2/l) I_2$ . <b>[1]</b> All of it threads the inner solenoid, so the flux linked with it is $N_1 \Phi = \mu_0 N_1 N_2 A I_2 / l$ . <b>[1]</b> By definition $N_1 \Phi = M I_2$ , so $\mathbf{M} = \mu_0 \mathbf{N}_1 \mathbf{N}_2 \mathbf{A} / l$ . The same expression follows if the current is put in the inner solenoid instead, which illustrates the <b>reciprocity</b> $M_{12} = M_{21}$ . <b>[1]</b>                                                                                                                                                                                                                                                                                                                                                               |
| 15      | Diagram: object between the lens and its first focus; a virtual, erect, magnified image at $D$ on the same side; $f$ , $u$ and $D$ labelled. <b>[1]</b> $\mathbf{M} = 1 + \mathbf{D}/f$ . <b>[1]</b> A smaller $f$ gives a <b>larger</b> magnifying power, since $M \propto 1/f$ ; but a very short focal length means a small, strongly curved lens with severe spherical and chromatic aberration, which is why a simple microscope is limited to about $\times 10$ in practice. <b>[1]</b>                                                                                                                                                                                                                                                                                                                                                                                             |
| 16      | $n = \sin[(A + \delta_m)/2] / \sin(A/2) = \sin 45^\circ / \sin 30^\circ = 0.7071/0.5 = 1.414 (= \sqrt{2})$ . <b>[2]</b> $v = c/n = (3.0 \times 10^8)/1.414 = 2.12 \times 10^8$ m s <sup>-1</sup> . <b>[1]</b> At minimum deviation the ray passes symmetrically, so $i = e = 45^\circ$ and $r_1 = r_2 = 30^\circ$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

| Q       | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 17      | Any three: (i) interference fringes are all of <b>equal width</b> , whereas diffraction bands become <b>narrower</b> away from the centre; (ii) interference maxima are of <b>equal intensity</b> , whereas the diffraction central maximum is far brighter than the rest, whose intensity falls off rapidly; (iii) interference arises from <b>two separate coherent wavefronts</b> , diffraction from the <b>secondary wavelets of one and the same wavefront</b> ; (iv) in interference the dark fringes are <b>perfectly dark</b> , in diffraction the minima are not; (v) the fringe width in interference depends on the slit <b>separation</b> $d$ , in diffraction on the slit <b>width</b> $a$ . |
| 17 (OR) | $a \sin \theta = \lambda$ gives $a = (5.0 \times 10^{-7})/0.5 = 1.0 \times 10^{-6} \text{ m} = \mathbf{1.0 \mu m}$ . <b>[2]</b> The angular width of the central maximum is $2\theta = \mathbf{60^\circ}$ (about 1.05 rad). <b>[1]</b> Note that the small-angle form $2\lambda/a = 1.0 \text{ rad} = 57^\circ$ is close but not exact here, because $30^\circ$ is not a small angle.                                                                                                                                                                                                                                                                                                                     |

## Section D

| Q       | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 18      | <b>(a)</b> Take a concentric spherical Gaussian surface of radius $r > R$ . By symmetry <b>E</b> is radial and of constant magnitude on it, so <b>E</b> $\parallel$ <b>dS</b> everywhere. <b>[1]</b> $\oint \mathbf{E} \cdot d\mathbf{S} = E \cdot 4\pi r^2 = q/\epsilon_0$ . <b>[1]</b> Hence <b>E</b> = $q/4\pi\epsilon_0 r^2$ - the shell behaves as though its whole charge were at its centre. For $r < R$ the enclosed charge is zero, so <b>E</b> = <b>0</b> everywhere inside. <b>[1]</b> <b>(b)</b> At 20 cm (outside): $E = (9 \times 10^9)(2.0 \times 10^{-6})/(0.20)^2 = \mathbf{4.5 \times 10^5 \text{ N C}^{-1}}$ , radially outward. At 5.0 cm (inside the shell): <b>E</b> = <b>0</b> .                                                                                                                                                          |
| 18 (OR) | <b>(a)</b> Take a coaxial cylinder of radius $r$ and length $l$ . By symmetry <b>E</b> is radial and constant over the curved surface; the flux through the two flat ends is <b>zero</b> because <b>E</b> $\perp$ to their normals there. <b>[1]</b> Charge enclosed = $\lambda l$ , so $E \cdot 2\pi r l = \lambda l/\epsilon_0$ . <b>[1]</b> Hence <b>E</b> = $\lambda/2\pi\epsilon_0 r$ , so $E \propto 1/r$ - not $1/r^2$ , because the symmetry is cylindrical. <b>[1]</b> <b>(b)</b> Series: $C = (3)(6)/9 = 2.0 \mu\text{F}$ , $U = \frac{1}{2}CV^2 = \frac{1}{2}(2.0 \times 10^{-6})(3600) = \mathbf{3.6 \times 10^{-3} \text{ J}}$ = 3.6 mJ. Parallel: $C = 9.0 \mu\text{F}$ , $U = \frac{1}{2}(9.0 \times 10^{-6})(3600) = \mathbf{1.62 \times 10^{-2} \text{ J}}$ = 16.2 mJ - four and a half times as much, since at a fixed voltage $U \propto C$ . |
| 19      | <b>(a)</b> Diagram: object AB beyond C, the ray parallel to the axis reflected through F, the ray through C returning on itself, image A'B' between C and F. <b>[1]</b> Triangles A'B'F and MPF are similar, so $A'B'/AB = B'F/PF$ ; triangles ABP and A'B'P are similar, so $A'B'/AB = B'P/BP$ . <b>[1]</b> Equating and putting in the sign convention ( $BP = -u$ , $B'P = -v$ , $PF = -f$ ) gives $(v-f)/f = v/u$ , which rearranges to $\mathbf{1/v + 1/u = 1/f}$ . <b>[1]</b> <b>(b)</b> $1/v = 1/(-20) - 1/(-30) = -1/60$ , so $v = \mathbf{-60 \text{ cm}}$ (60 cm in front of the mirror). $m = -v/u = -2$ , so the image is <b>8.0 cm</b> high, <b>inverted, real and magnified</b> - as expected for an object between F and C.                                                                                                                       |
| 19 (OR) | <b>(a)</b> For the first lens, $1/v_1 - 1/u = 1/f_1$ . <b>[1]</b> The image $v_1$ acts as the <b>object</b> for the second lens (the lenses being thin and in contact, the separation is negligible), so $1/v - 1/v_1 = 1/f_2$ . <b>[1]</b> Adding, $v_1$ cancels: $1/v - 1/u = 1/f_1 + 1/f_2$ , so $\mathbf{1/F = 1/f_1 + 1/f_2}$ , i.e. $P = P_1 + P_2$ . <b>[1]</b> <b>(b)</b> $1/F = 1/30 + 1/(-20) = -1/60$ , so <b>F</b> = <b>-60 cm</b> : the combination is <b>diverging</b> ( $P = 3.33 - 5.0 = -1.67 \text{ D}$ ), because the concave lens is the stronger of the two.                                                                                                                                                                                                                                                                                |
| 20      | (i) So that each appliance receives the <b>full 220 V</b> it is rated for, and so that each can be switched on or off <b>independently</b> without breaking the others' circuit. (ii) $I = P/V = 2000/220 = \mathbf{9.1 \text{ A}}$ . (iii) $R = V^2/P = (220)^2/2000 = \mathbf{24.2 \Omega}$ ( $= V/I$ ). (iv) <b>No</b> . The normal working current, 9.1 A, already exceeds 5 A, so the fuse would melt as soon as the iron was switched on; a fuse of about 15 A is needed. (v) So that when it blows, the appliance is disconnected from the <b>high potential</b> side. A fuse in the neutral wire would break the circuit but leave the appliance still connected to the live wire, so its body could stay at 220 V and give a fatal shock.                                                                                                               |

## Mark distribution over the First Term chapters

| Chapter                                           | A         | B         | C         | D         | Total     |
|---------------------------------------------------|-----------|-----------|-----------|-----------|-----------|
| Electrostatics                                    | 2         | 2         | —         | 5         | <b>9</b>  |
| Current Electricity                               | 3         | 2         | 3         | 5         | <b>13</b> |
| Magnetic Effect of Current and Magnetism          | 2         | 2         | 6         | —         | <b>10</b> |
| Electromagnetic Induction and Alternating Current | 2         | 2         | 6         | —         | <b>10</b> |
| Electromagnetic Waves                             | 1         | 2         | —         | —         | <b>3</b>  |
| Optics — Ray Optics and Optical Instruments       | 2         | 2         | 6         | 5         | <b>15</b> |
| Optics — Wave Optics                              | 2         | 2         | 6         | —         | <b>10</b> |
| <b>Total</b>                                      | <b>14</b> | <b>14</b> | <b>27</b> | <b>15</b> | <b>70</b> |